Exploratory Data Analysis of a Used Smartphone Resale Market Dataset

A Data Cleaning and Visual Analysis Project

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Dataset Source	used_phones_uncleaned_dataset.csv (self-compiled resale listings)

1. Objective

This project performs an exploratory data analysis (EDA) on a dataset of second-hand smartphone listings collected from an online resale marketplace. The goal is to understand how a phone's brand, age, usage, RAM, and battery condition relate to its resale price, and to prepare the raw data for any future modelling work by cleaning it thoroughly.

The analysis focuses on the following questions:

- How is the resale price distributed across the dataset, and which brands dominate the listings?
- How much missing data and duplication exists, and how should it be handled?
- Which numerical features (RAM, usage hours, battery backup) show a relationship with price?
- Do categorical attributes such as battery health, charger availability, or city show any visible pattern with price?
- What data quality issues (inconsistent casing, stray whitespace) need correction before the data can be trusted?

2. Dataset Overview

The raw file contains **208 rows** and **12 columns**, each row representing one used-phone listing. After removing exact duplicate rows, **203 unique listings** remain.

Feature descriptions

Column	Type	Description
Phone_ID	int	Unique identifier for each listing
Brand	text	Manufacturer (e.g. Samsung, Apple, Xiaomi)
Model	text	Specific phone model
Release_Year	int	Year the model was originally released
Usage_Hours	int	Estimated total hours of prior use
Battery_Health	text	Reported battery condition: Good / Average / Poor
Charger_Included	text	Whether the original charger is included: Yes / No
Owner	int	Number of previous owners (1-3)
RAM_GB	float	RAM capacity in gigabytes
Battery_Backup_Hrs	float	Battery backup on a full charge, in hours
Price_USD	int	Listed resale price in US dollars
City	text	City where the listing is posted

Sample records

ID	Brand	Model	Year	RAM	Battery Hrs	Price
1	Oneplus	Nord Ce	2019	6.0	12.9	$790
2	Oneplus	Nord Ce	2018	12.0	10.5	$280
3	Xiaomi	Poco X3	2017	6.0	4.3	$656
4	Samsung	Galaxy M31	2016	6.0	7.0	$169
5	Oneplus	Nord 2	2017	6.0	8.2	$484

3. Data Quality and Cleaning

Missing values. Six columns contained missing entries in the raw file:

Column	Missing values
Battery_Health	3
Charger_Included	6
RAM_GB	6
Battery_Backup_Hrs	7
City	7
All other columns	0

The missing values are limited to a small fraction of rows (under 4% per column) and are spread across both numerical and categorical fields. Rather than dropping these rows outright, they were left in place for the visual analysis, since seaborn and pandas handle missing values gracefully in histograms and correlation calculations. A row-removal step (`dropna()`) was written but intentionally left commented out, since removing ~9% of the dataset was judged to cost more information than the missing cells themselves.

Duplicates. The dataset contained **5 fully duplicate rows** (identical values across every column). These were removed with `drop_duplicates()`, bringing the working dataset down from 208 to **203 rows**.

Outliers. The numerical ranges were inspected using `describe()`. Usage_Hours spans 517 to 19,982 hours and Price_USD spans $89 to $898 - both wide but plausible for a resale-listing dataset, so no values were treated as outliers or removed.

Type and text corrections. Brand, Model, Battery_Health, Charger_Included, and City are text columns. Inspecting their unique values revealed inconsistent capitalisation (for example `oneplus` and `Oneplus` being treated as different categories) and stray leading/trailing spaces. Every text column was corrected with `.str.strip().str.title()`, which trims whitespace and standardises capitalisation so that each brand and city is represented by exactly one category.

4. Descriptive Analysis

Summary statistics - numerical columns

Statistic	Release Year	Usage Hrs	RAM (GB)	Battery Hrs	Price (USD)
Mean	2019.3	10,342.6	7.20	9.27	491.31
Std Dev	2.29	5,868.8	2.81	3.24	234.88
Min	2016	517	4.0	4.0	89
25%	2017	4,965	4.0	6.57	302.5
Median	2019	9,834	6.0	9.00	489
75%	2021	15,449	8.0	11.83	685
Max	2023	19,982	12.0	15.0	898

Observations on categorical variables

- Oppo is the most listed brand (35 listings), closely followed by the other six brands, so no single manufacturer dominates the marketplace overwhelmingly.
- Battery_Health splits fairly evenly across Good (74), Average (65) and Poor (61), suggesting sellers are reporting a realistic range of conditions rather than only listing well-kept phones.
- Ownership is nearly balanced between 1, 2 and 3 previous owners (70 / 69 / 64), indicating the dataset covers phones at different points in their resale life, not just first-hand resales.
- Chennai has the largest share of listings among the seven cities, though the spread across cities is fairly even overall.

5. Visual EDA

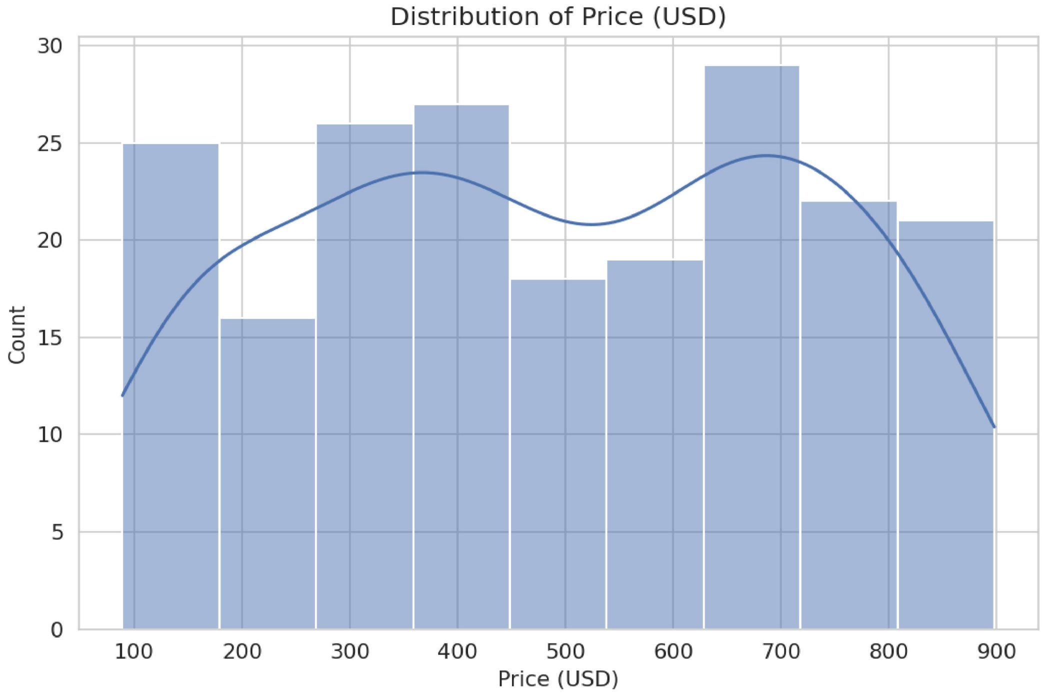

Figure 1. Distribution of Price (USD)

Observation: Resale prices are fairly spread out between roughly $90 and $900, with a broad hump around $400-$600 rather than a sharp peak, meaning the dataset covers budget and premium listings in comparable numbers.

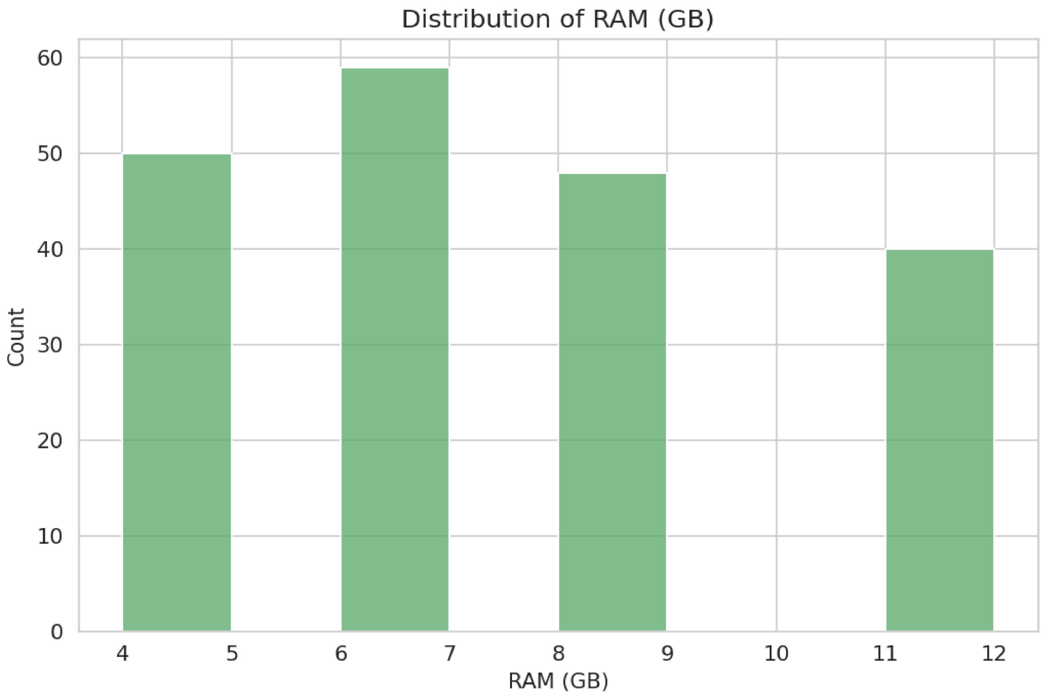

Figure 2. Distribution of RAM (GB)

Observation: Most listed phones cluster at 4, 6 and 8 GB of RAM, with 12 GB devices forming a smaller minority - consistent with mid-range phones being the most commonly resold segment.

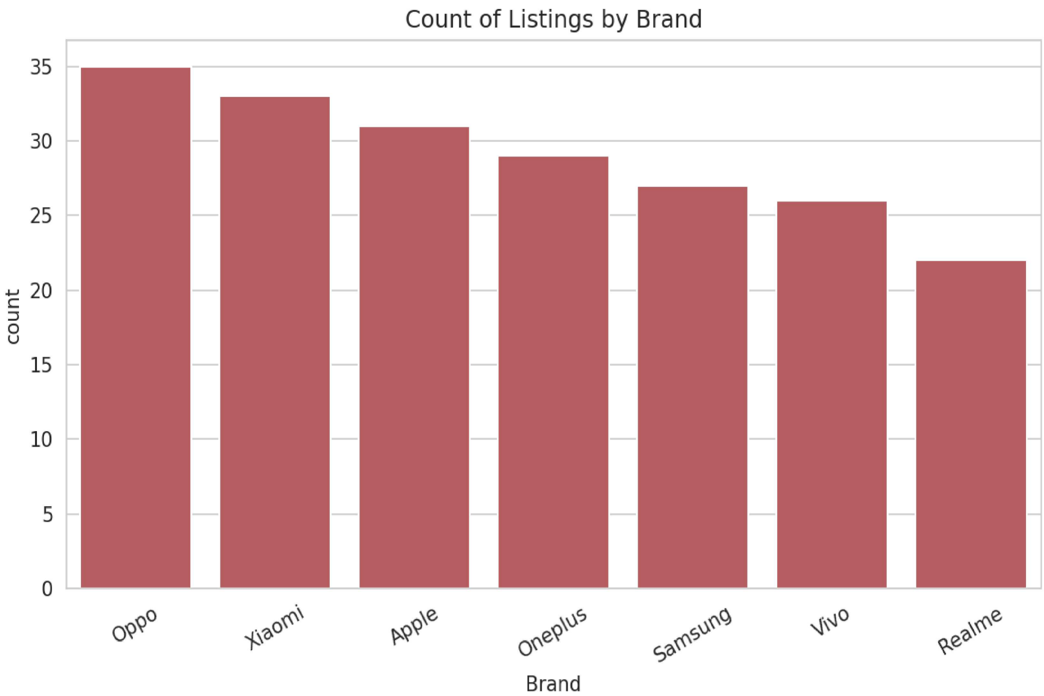

Figure 3. Count of Listings by Brand

Observation: Listing counts are close across all seven brands (roughly 25-35 each), so brand popularity in this dataset is fairly balanced rather than skewed toward one manufacturer.

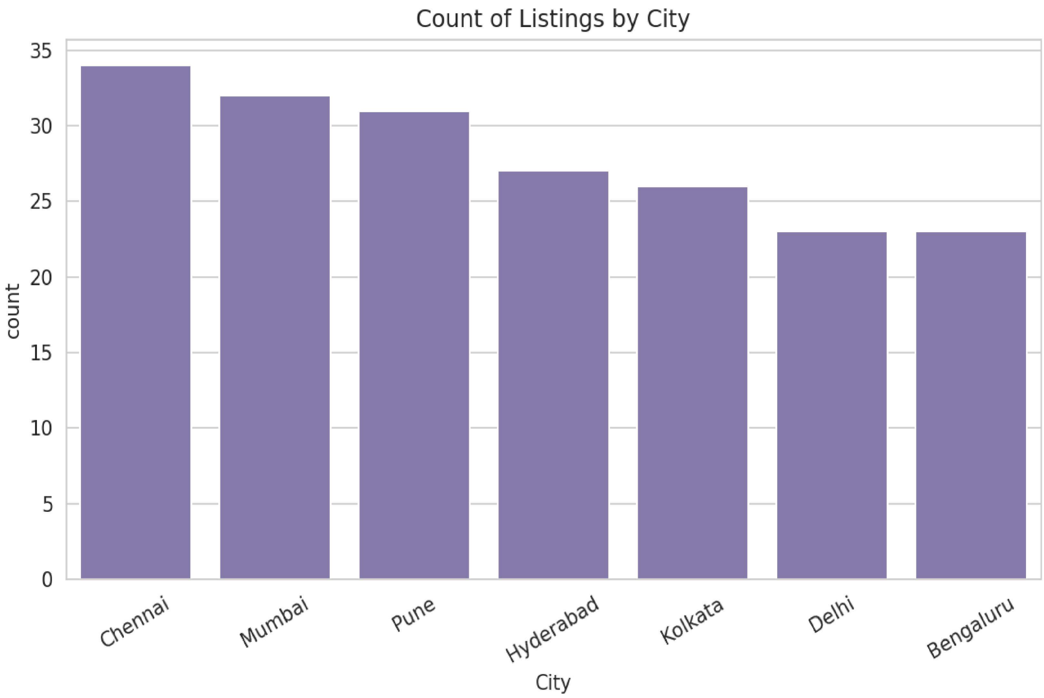

Figure 4. Count of Listings by City

Observation: Listings are distributed across all seven cities with only mild variation, showing the marketplace draws sellers fairly evenly from each region rather than one city dominating supply.

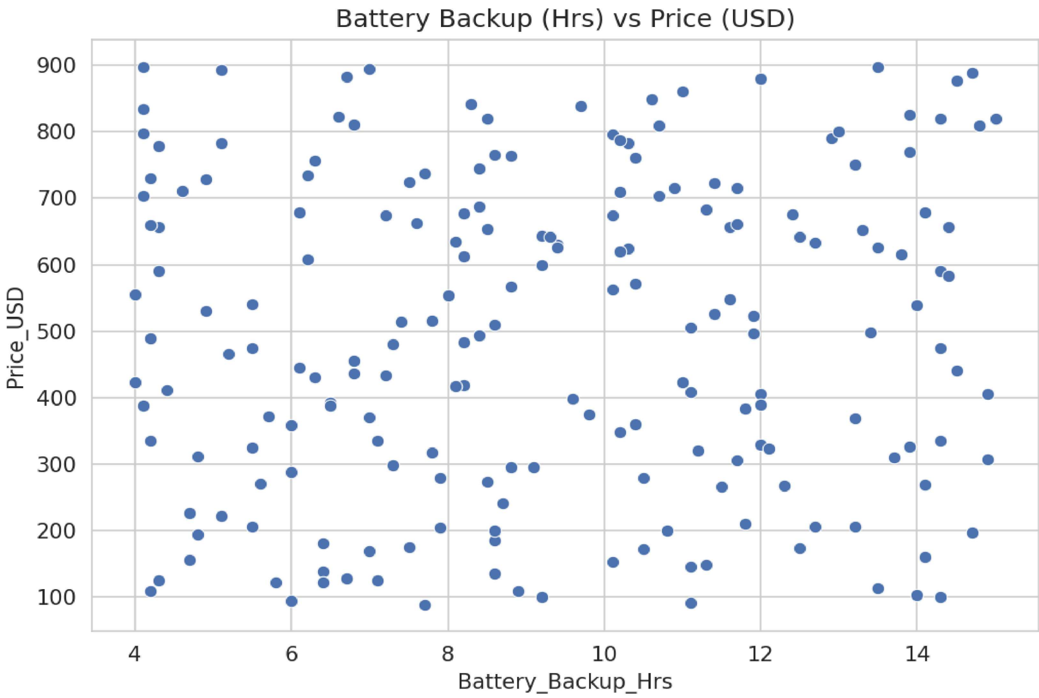

Figure 5. Battery Backup (Hrs) vs Price (USD)

Observation: Points are scattered with no visible upward or downward trend, suggesting battery backup alone is not a strong driver of resale price in this dataset.

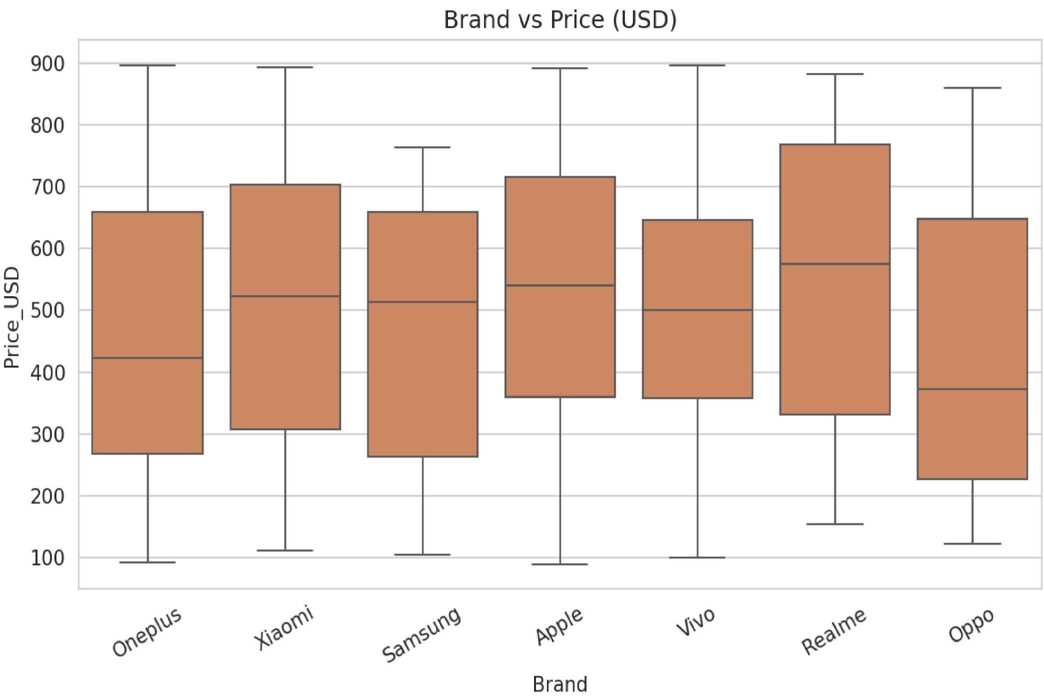

Figure 6. Brand vs Price (USD)

Observation: Median prices and spreads are broadly similar across brands, with a few brands showing wider interquartile ranges - indicating brand alone does not cleanly separate low- and high-priced listings; model and condition likely matter more.

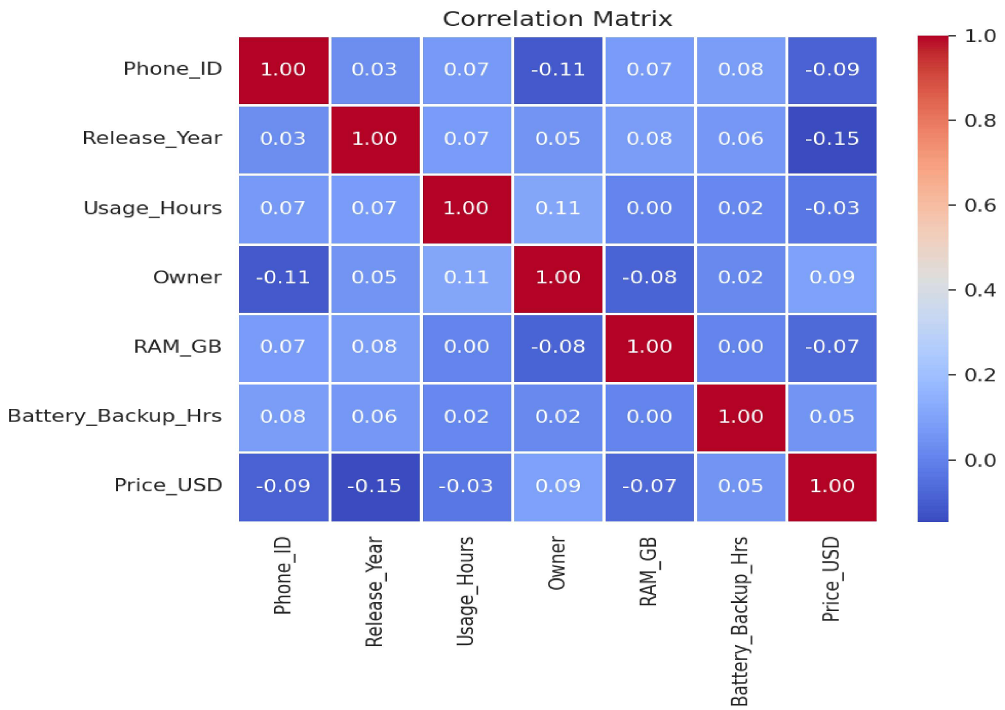

Figure 7. Correlation Matrix of Numerical Features

Observation: Correlations between the numerical features and Price_USD are all close to zero, the strongest being Battery_Backup_Hrs at about 0.05. This indicates that price in this dataset is not linearly explained by any single numerical feature and likely depends more on categorical factors (brand, model, condition) or on factors not captured in the data.

6. Key Findings

- Price is not strongly correlated with any single numerical feature: RAM, usage hours, and battery backup all show near-zero correlation with Price_USD, with battery backup the highest at only 0.05.
- The dataset is well balanced across categorical groups - brand, city, battery health and owner count are all spread fairly evenly, which reduces the risk of one dominant category skewing later analysis.
- Data quality issues were minor but real: 5 duplicate rows (about 2.4% of the raw data) and inconsistent text casing in brand and city names had to be corrected before the categories could be trusted for grouping or counting.
- Missing values were concentrated in five columns and never exceeded 7 rows (about 3.4%) in any single column, so they were left in place rather than dropped, preserving more of the dataset for analysis.
- Median resale price ($489) sits close to the mean ($491), indicating the price distribution is roughly symmetric rather than skewed by a small number of very expensive or very cheap listings.

Limitations of the dataset

- The dataset covers only 203 unique listings after deduplication - a modest sample size that limits how confidently the observed patterns generalise to the wider resale market.
- Condition-related fields (Battery_Health) are self-reported categories rather than measured values, which can introduce inconsistency between sellers.
- No feature captures screen condition, storage capacity, or physical damage, all of which typically influence resale price and may explain why price does not correlate strongly with the available numerical features.
- The data covers a single snapshot in time and does not capture how prices change as a listing ages on the marketplace.

7. Conclusion

This project walked through a complete exploratory data analysis workflow: loading the raw listing data, checking its structure and quality, handling duplicates and missing values, and visualising both numerical and categorical features to look for patterns tied to resale price. The most useful outcome was a working, reusable cleaning routine - deduplication plus consistent text formatting - and a clear picture of which relationships in the data are and are not strong.

The main challenge was that none of the numerical features showed a meaningful correlation with price, which made the univariate and bivariate charts more useful for understanding the shape of the data than for predicting price directly. This reinforced the importance of visual inspection before jumping to any modelling step, since a quick correlation check alone would have suggested (incorrectly) that the dataset held little signal at all.

For future analysis, it would be worth engineering additional features - such as phone age (current year minus Release_Year) or a combined condition score from Battery_Health and Owner count - and testing whether categorical variables like Brand and Model, when properly encoded, explain more of the variation in price than the numerical columns did on their own.